

Assignment 5

Due date: 2/26

Problem 1 (10 points) Consider the following confusion matrix.

actual class	predicted class		
		C1 (positive)	C2 (negative)
	C1 (positive)	132	29
	C2 (negative)	87	218

Compute *sensitivity*, *specificity*, *precision*, *accuracy*, *F-measure*, *F2*, and *MCC* measures. You must do all calculations yourself and must show all your calculation steps.

Problem 2 (10 points) Suppose you built two classifier models $M1$ and $M2$ from the same training dataset and tested them on the same test dataset using 10-fold cross-validation. The error rates obtained over 10 iterations (in each iteration the same training and test partitions were used for both $M1$ and $M2$) are given in the table below. Determine whether there is a significant difference between the two models using the t-test that we discussed in the class. Use a significance level of 1%. If there is a significant difference, which one is better?

Iteration	M1	M2
1	0.12	0.08
2	0.08	0.11
3	0.10	0.12
4	0.13	0.05
5	0.05	0.09
6	0.09	0.13
7	0.10	0.19
8	0.08	0.04
9	0.25	0.20
10	0.19	0.12

Note: When you calculate $\text{var}(M1 - M2)$, calculate a sample variance (not a population variance).

You must do all calculations yourself and must show all calculation steps.

Problem 3 (10 points). The following table shows a test result of a classifier on a dataset.

Tuple id	Actual Class	Probability
1	P	0.75
2	N	0.32
3	N	0.87
4	P	0.93
5	P	0.84
6	P	0.48
7	N	0.95
8	P	0.59
9	N	0.63
10	P	0.74

- (1). For each row, compute TP , FP , TN , FN , TPR , and FPR .
- (2). Plot the ROC curve for the dataset. You must draw the curve yourself (i.e., don't use R or any other software to generate the curve).

Problem 4 (10 points). This problem is a small experiment of handling an unbalanced dataset for classification. Use *hw5_p4_train.csv* and *hw5_p4_test.csv* files. You must use the *rpart* algorithm in R to build decision trees.

- (1). Build a decision tree model from *hw5_p4_train.csv* and test it on *hw5_p4_test.csv*. Include the resulting confusion matrix in your submission.
- (2). Create an undersampled training dataset from *hw5_p4_train.csv* and name it *hw5_p4_train_undersampled.csv*. Build a decision tree model from this undersampled dataset and test it on *hw5_p4_test.csv*. Include the resulting confusion matrix in your submission.
- (3). Create an oversampled training dataset from *hw5_p4_train.csv* and name it *hw5_p4_train_oversampled.csv*. Build a decision tree model from this oversampled dataset and test it on *hw5_p4_test.csv*. Include the resulting confusion matrix in your submission.
- (4). What conclusion can you draw from this experiment?

Note: You may use any methods to create undersampled and oversampled training datasets. You must include in your submission name(s) of the methods/functions you used and brief description of these methods/functions.

Submission:

Include all files in a single archive file and name it *LastName_FirstName_HW5.EXT*. Here, "EXT" is an appropriate file extension (e.g., *zip* or *rar*).